

Cornell Bowers C-IS
College of Computing and Information Science

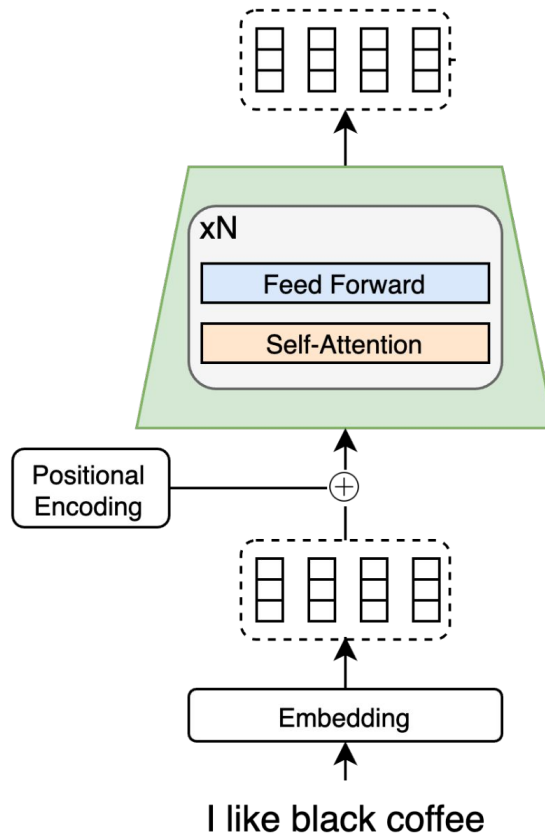
Deep Learning

Week 05: ViTs and CLIP

Thanks to

Varsha Kishore
Justin Lovelace
Anissa Dallmann
Dylan Van Bramer

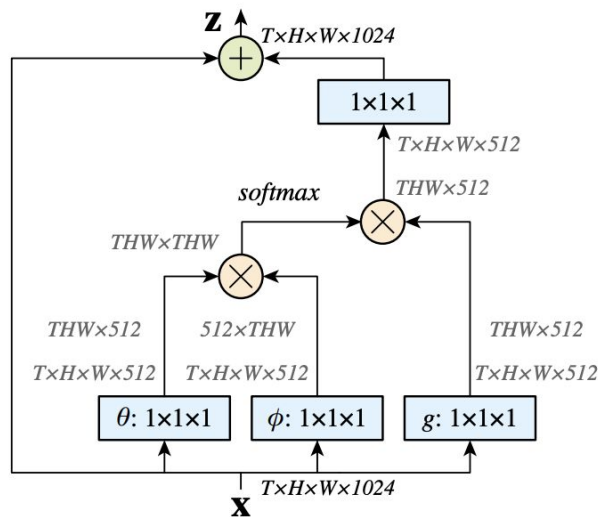
Remember Transformers?



How can we use Transformers on Images?

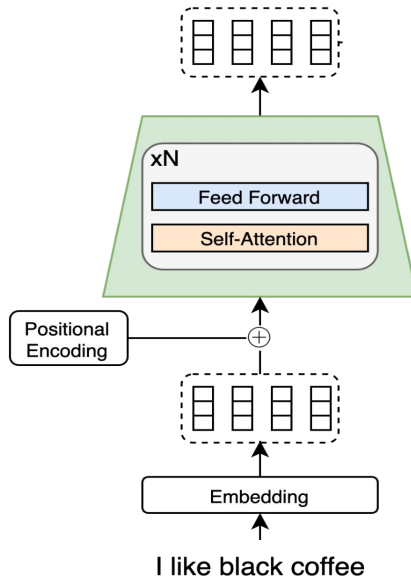
How to use Attention for Vision Tasks?

Attempt #1: Add attention to existing CNNs



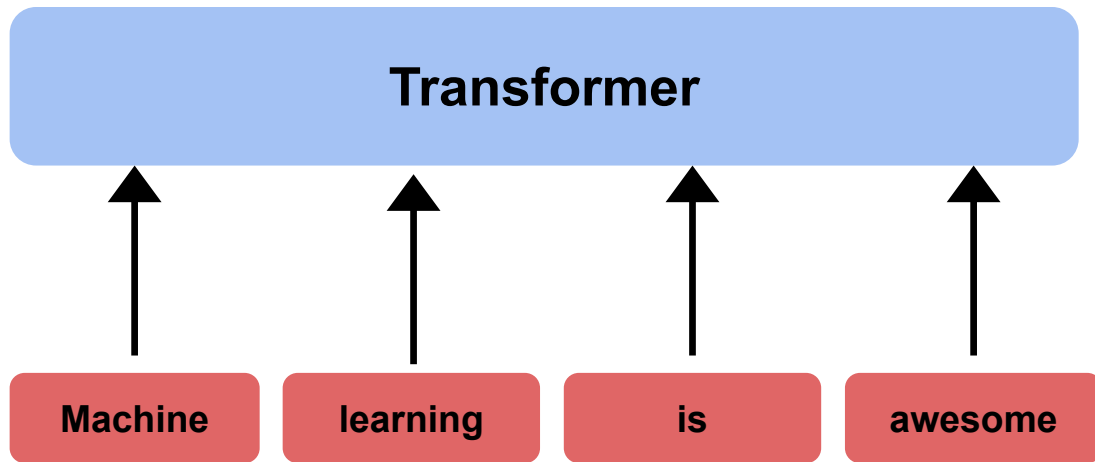
How to use Attention for Vision Tasks?

Attempt #2: Adapt standard transformers to image data

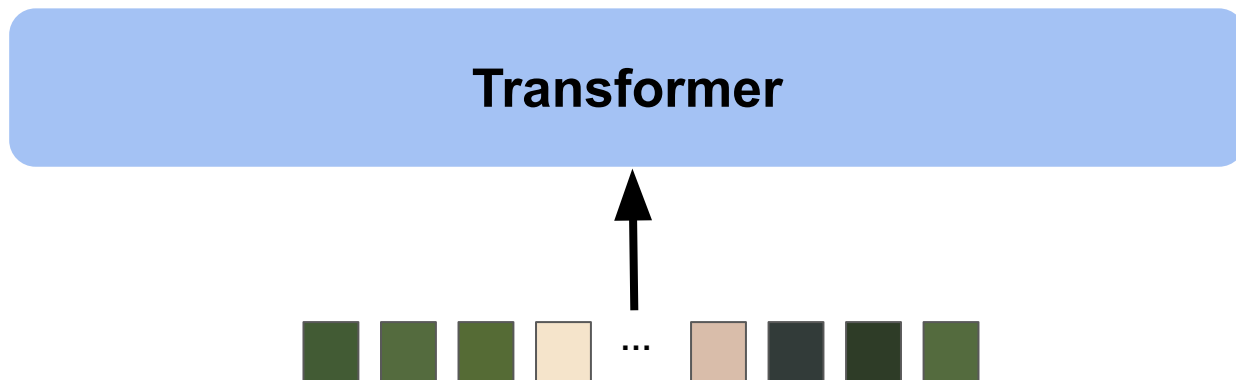


Can we extend this
idea to images?

Machine learning
is awesome



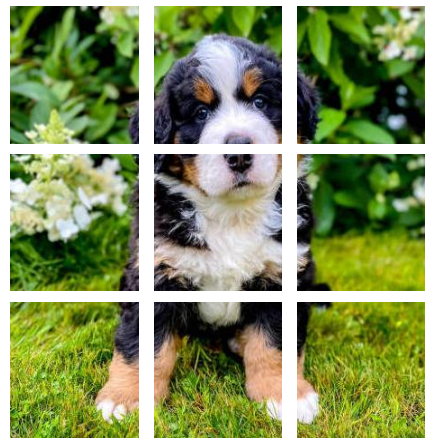
Idea1: Use pixels as input tokens



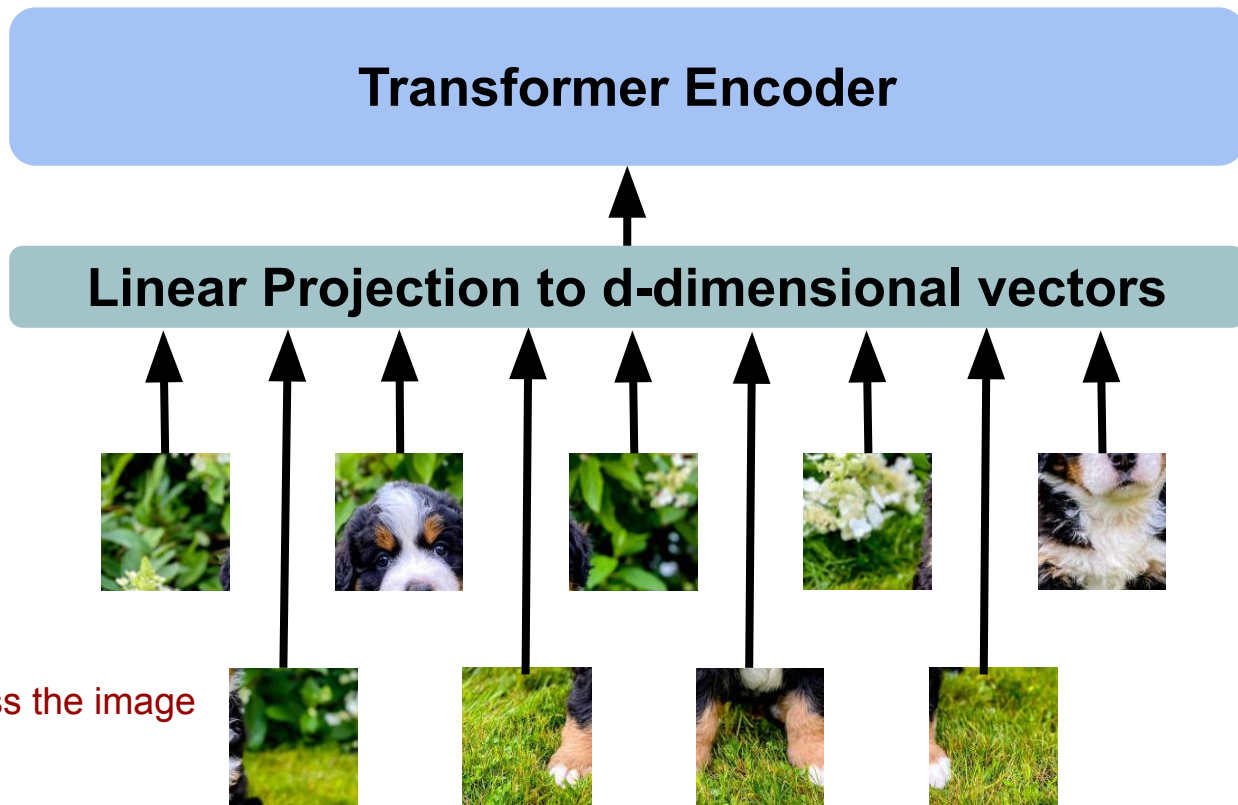
Discuss:

- How big would the attention matrix be for an $R \times R$ image?
- How much memory do we need for a 128×128 image, with 16 attention heads, and 48 layers - for attention alone?
- What if the image was 32×32 pixels?

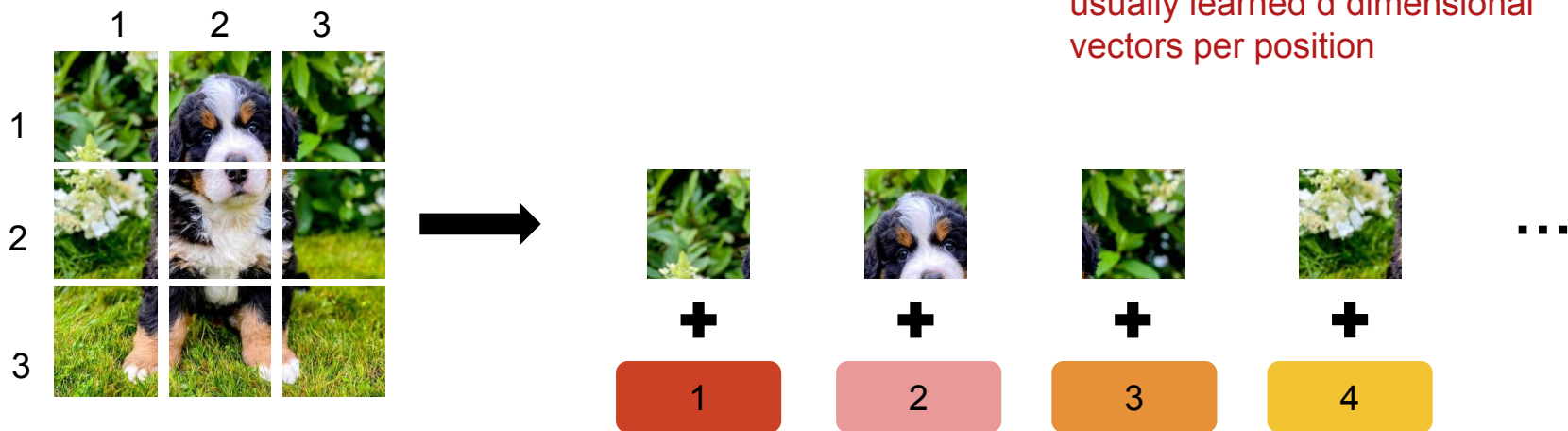
Idea2: use image patches as input tokens



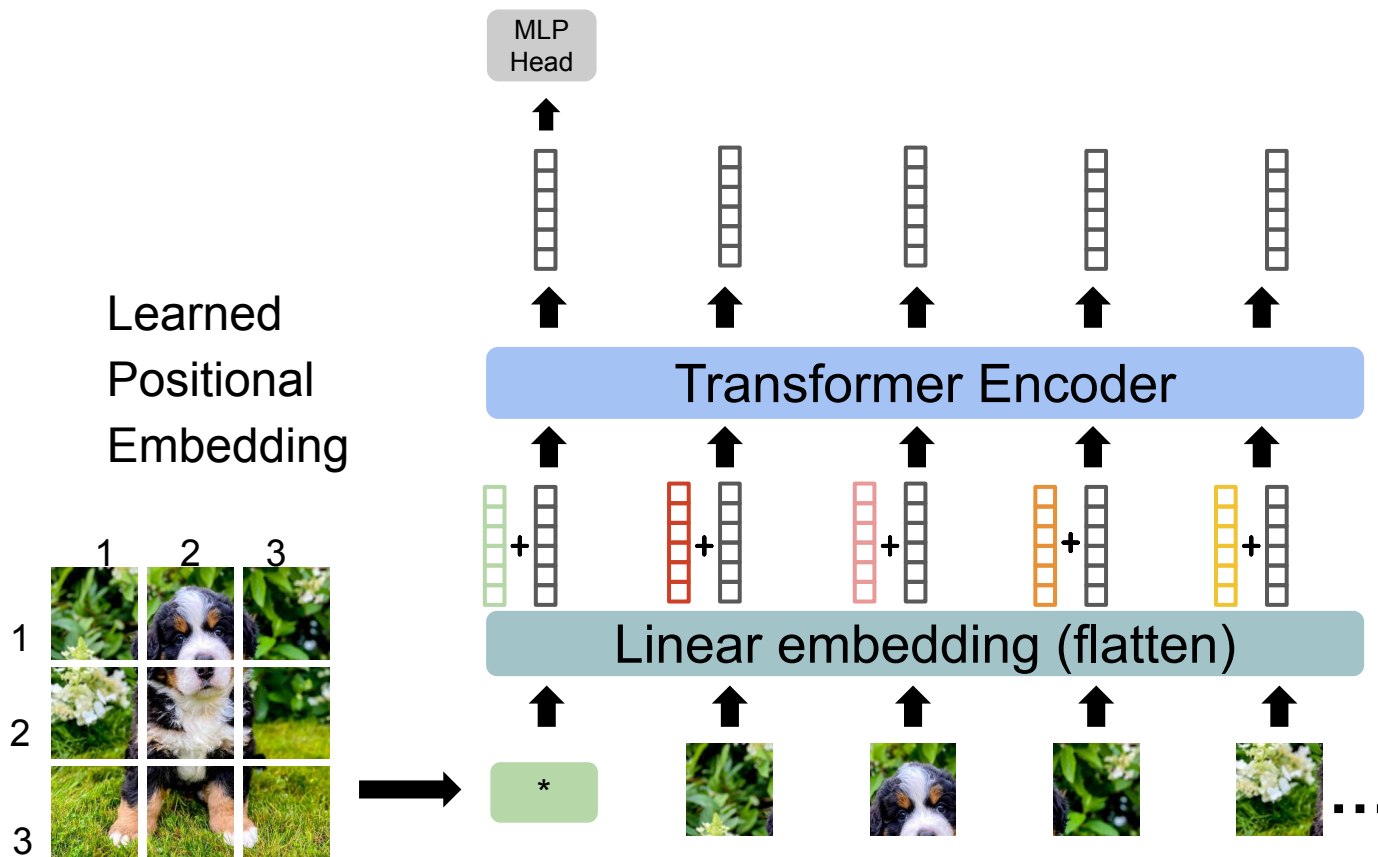
CNNs can be used to preprocess the image before converting it to patches!



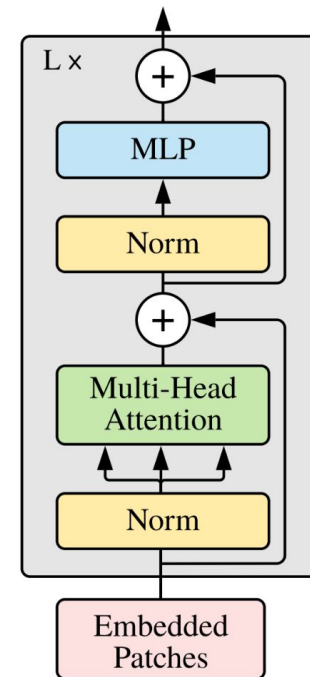
Adding positional embeddings



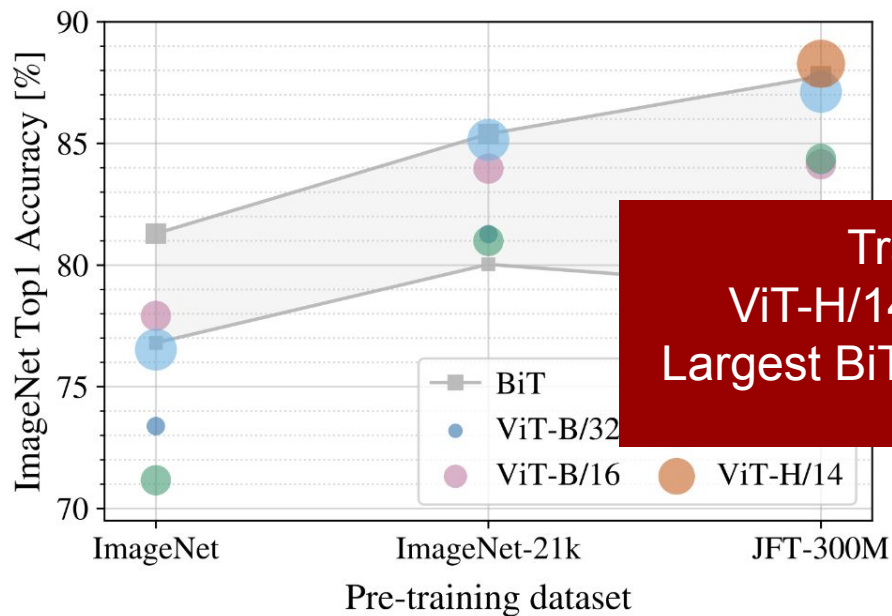
Vision Transformer



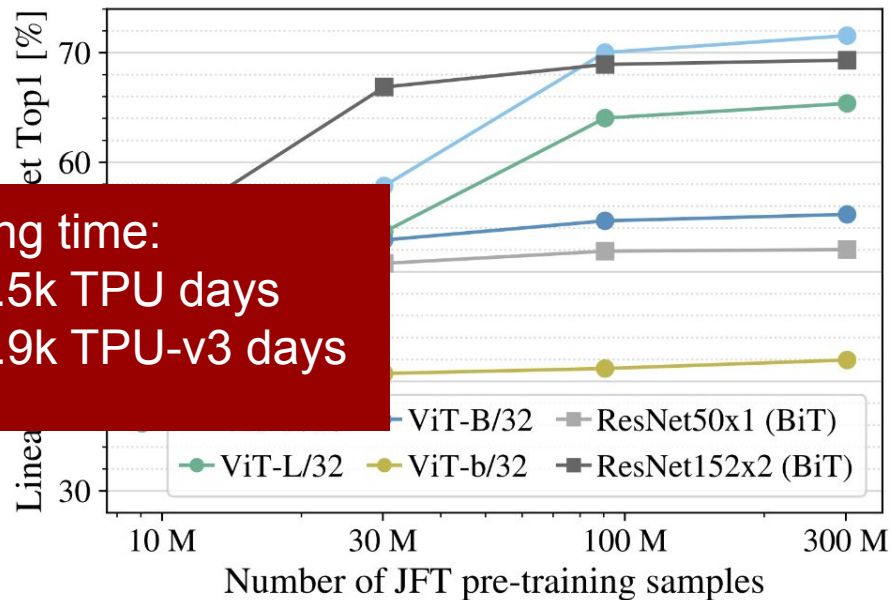
Transformer Encoder



ViT Results



Training time:
ViT-H/14: 2.5k TPU days
Largest BiT: 9.9k TPU-v3 days



ViT Summary

Model:

- Model is almost identical to Transformers for text sequences
- Replace words with $P \times P$ pixel image patches, $P \in \{14, 16, 32\}$ (no overlap)
- Each patch is embedded linearly into a vector of size 1024
- 1D **learned** positional embeddings

Training:

- For pre-training, optimize for image classification on large supervised dataset (e.g. ImageNet 21K, JFT -300M)
- For fine-tuning, learn a new classification head on a small dataset (e.g. CIFAR-100)

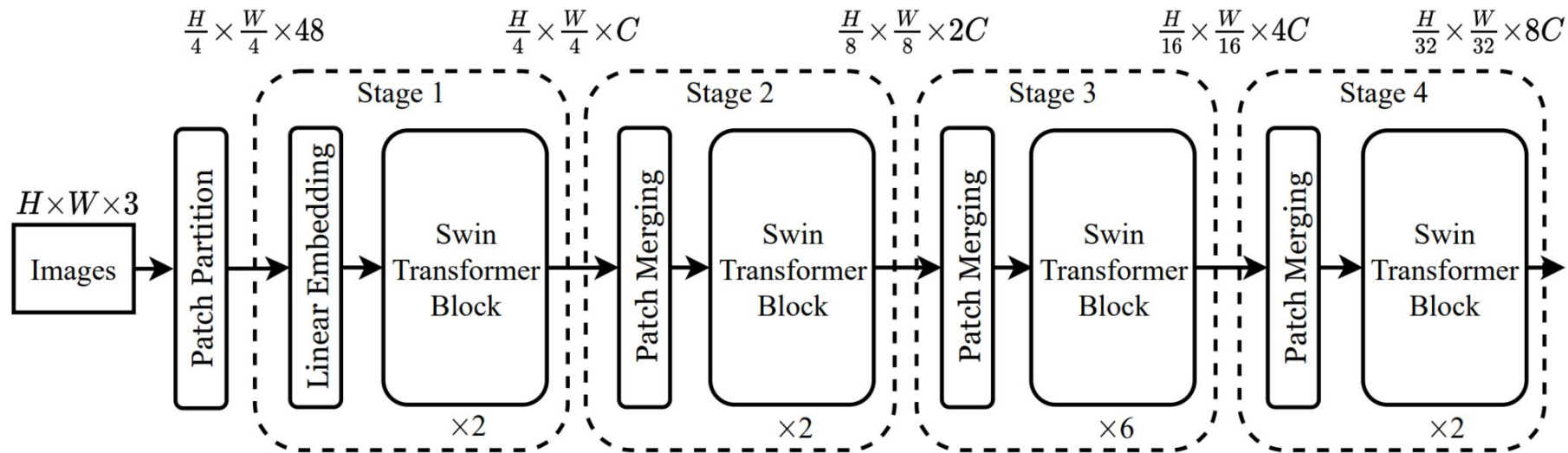
ACTIVITY: When do ViTs outperform CNNs, and vice versa?

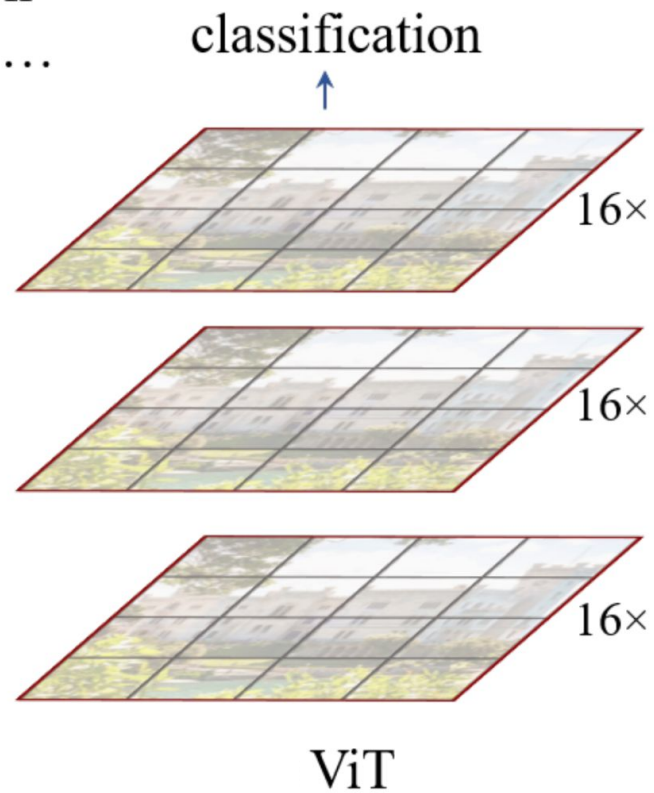
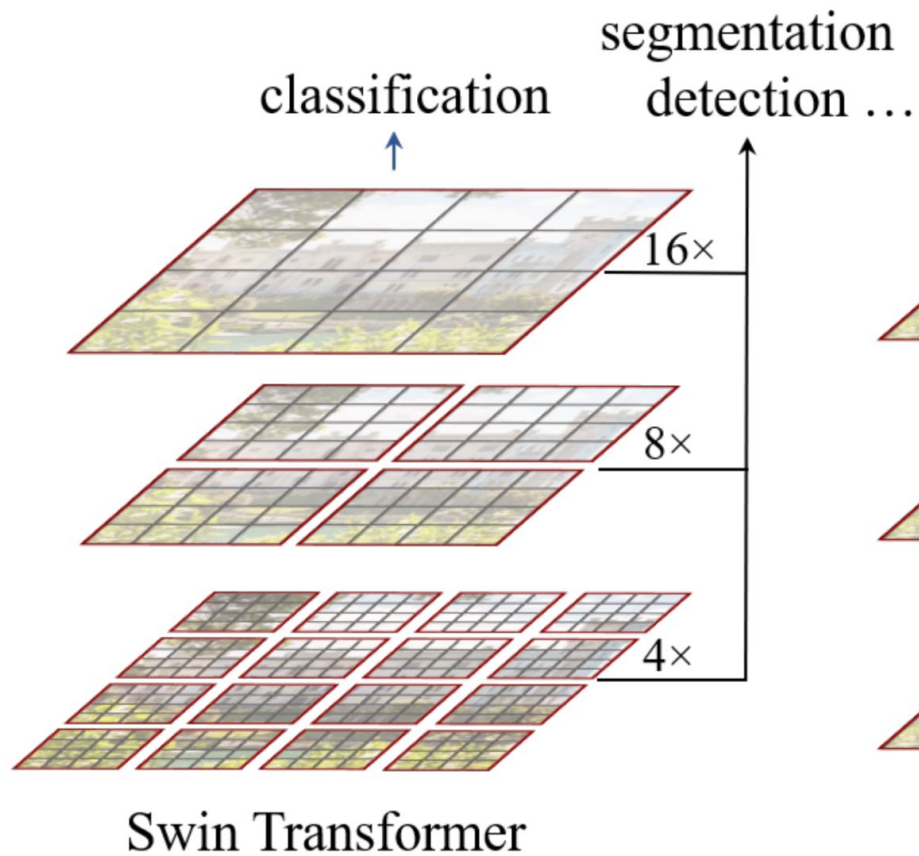
Think about what you know about transformers -
what are some of their drawbacks?

When is it “worth it” to use transformers instead of
just CNNs?

Idea 3: Swim Transformers

Hierarchical architecture that has the flexibility to model at various scales

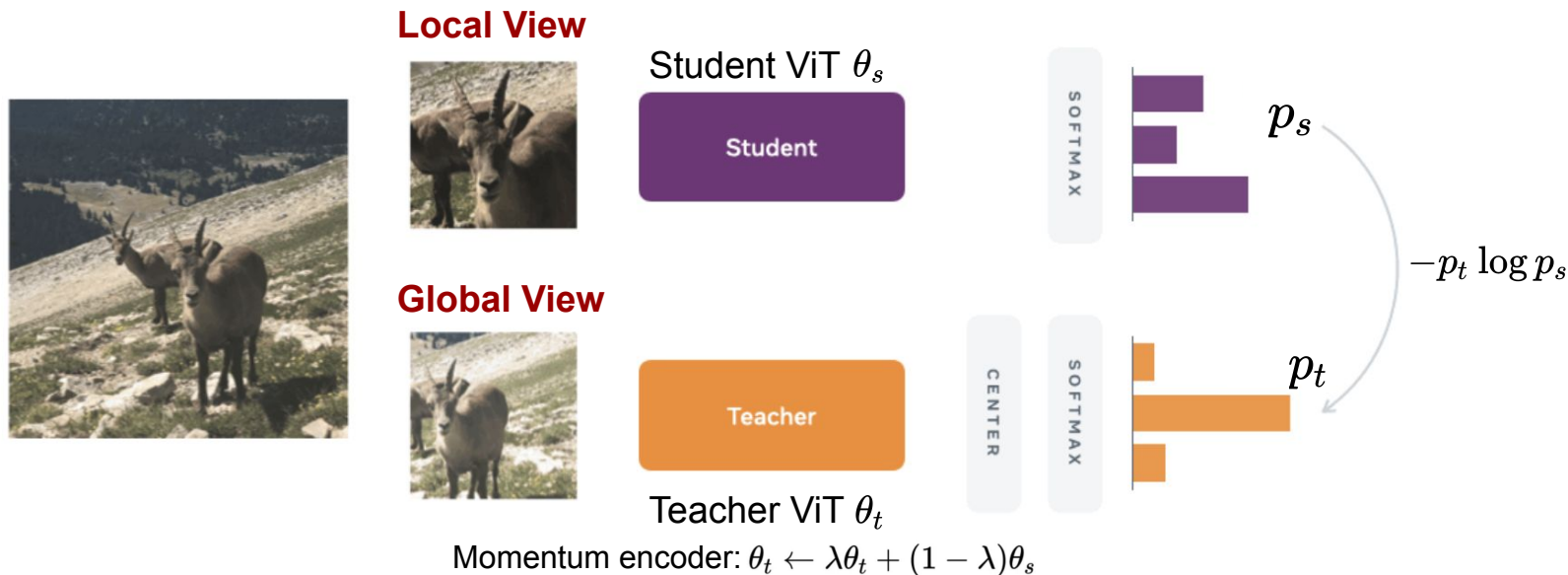




Performance

method	image size	#param.	FLOPs	throughput (image / s)	ImageNet top-1 acc.
R-101x3 [38]	384^2	388M	204.6G	-	84.4
R-152x4 [38]	480^2	937M	840.5G	-	85.4
ViT-B/16 [20]	384^2	86M	55.4G	85.9	84.0
ViT-L/16 [20]	384^2	307M	190.7G	27.3	85.2
Swin-B	224^2	88M	15.4G	278.1	85.2
Swin-B	384^2	88M	47.0G	84.7	86.4
Swin-L	384^2	197M	103.9G	42.1	87.3

Self-Supervised Vision Transformers (DiNO)



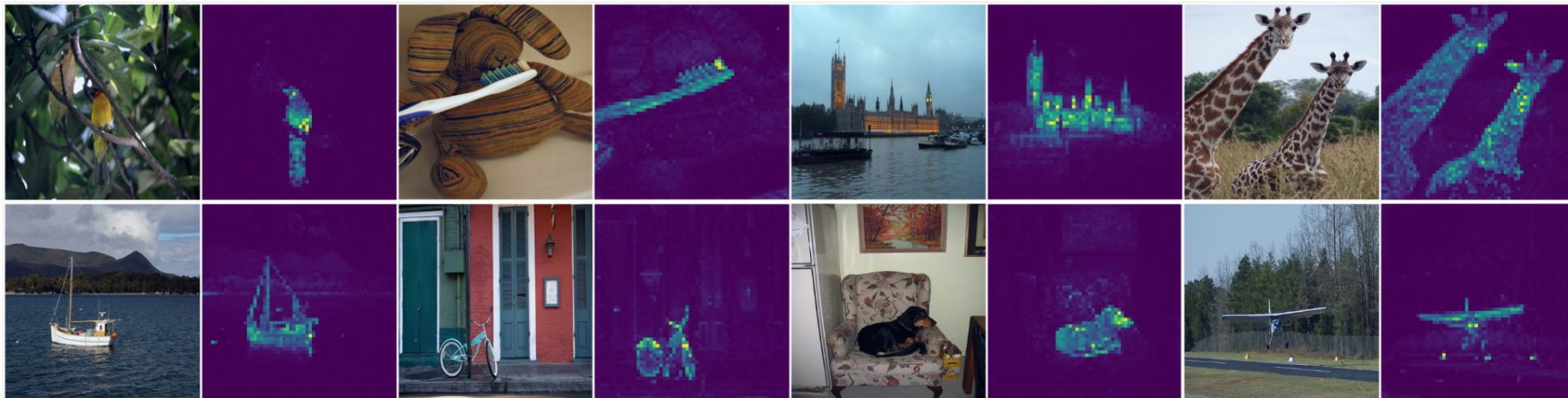
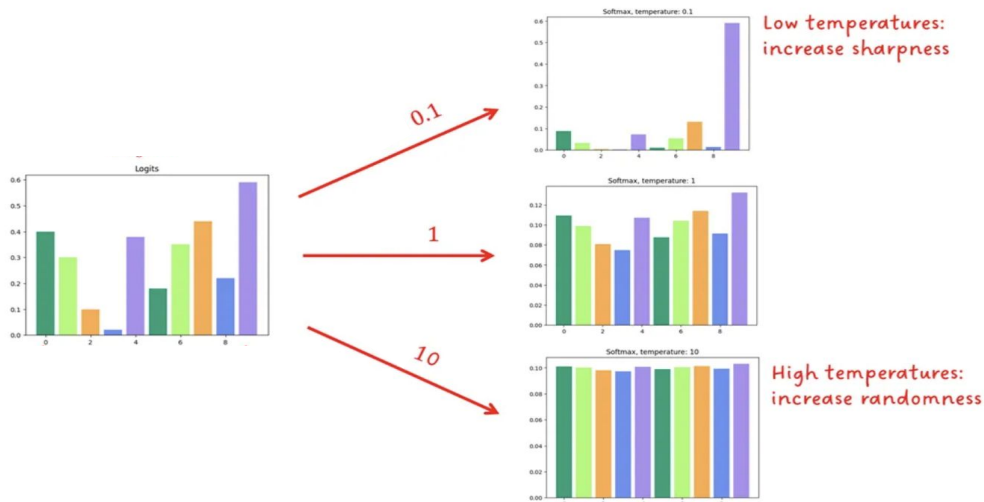


Figure 1: **Self-attention from a Vision Transformer with 8×8 patches trained with no supervision.** We look at the self-attention of the [CLS] token on the heads of the last layer. This token is not attached to any label nor supervision. These maps show that the model automatically learns class-specific features leading to unsupervised object segmentations.

Centering and sharpening

- Centering prevents one dimension from dominating
- Sharpening prevents learning a uniform distribution

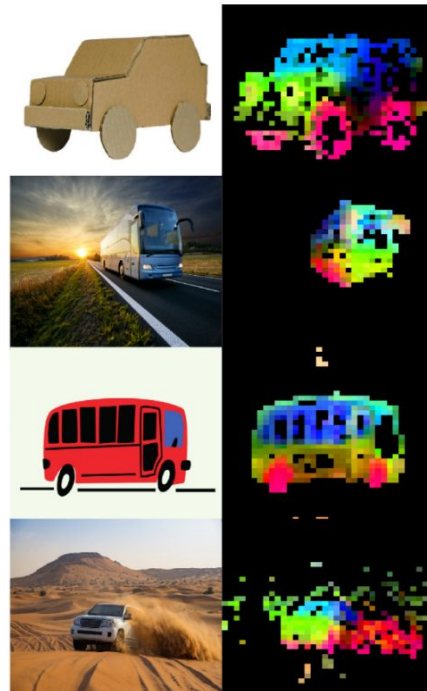
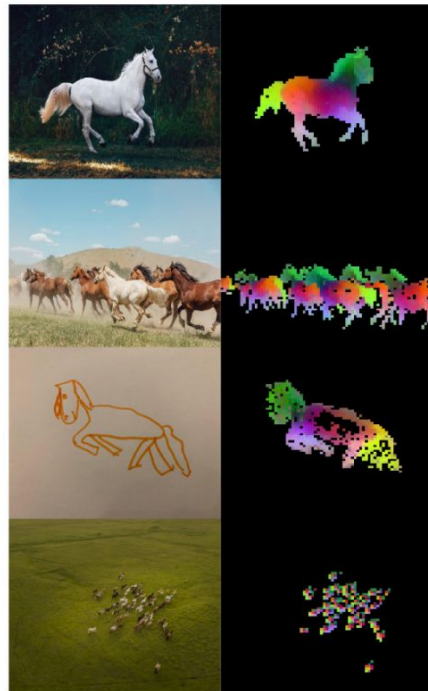
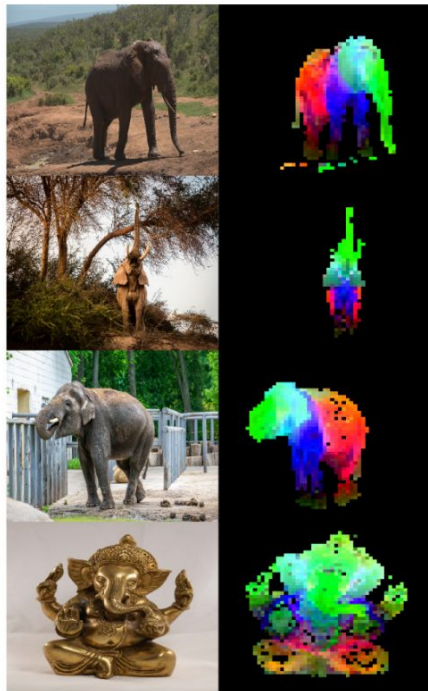
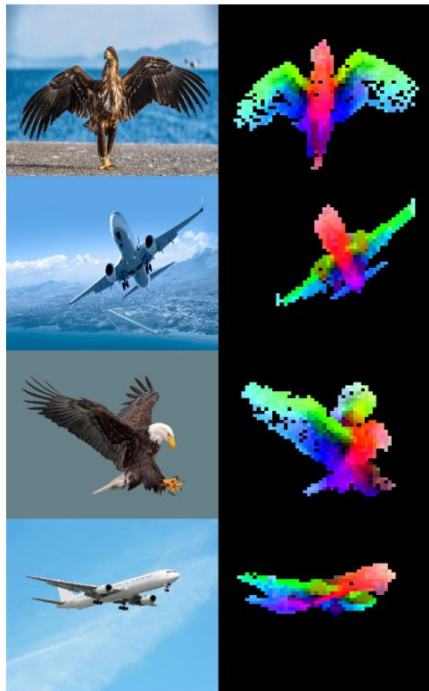
$$\text{softmax}(x)_i = \frac{e^{\frac{y_i}{T}}}{\sum_j^N e^{\frac{y_j}{T}}}$$



DINO v2

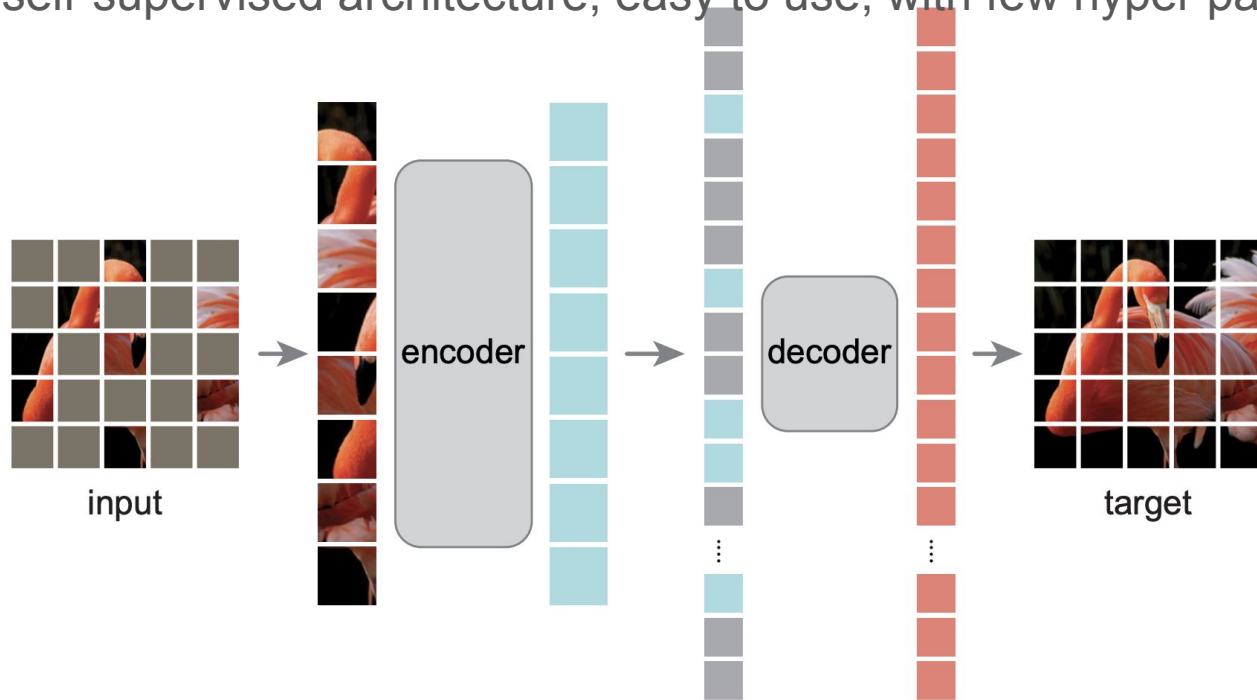
	INet-1k k-NN	INet-1k linear
iBOT	72.9	82.3
+ (our reproduction)	74.5 $\uparrow 1.6$	83.2 $\uparrow 0.9$
+ LayerScale, Stochastic Depth	75.4 $\uparrow 0.9$	82.0 $\downarrow 1.2$
+ 128k prototypes	76.6 $\uparrow 1.2$	81.9 $\downarrow 0.1$
+ KoLeo	78.9 $\uparrow 2.3$	82.5 $\uparrow 0.6$
+ SwiGLU FFN	78.7 $\downarrow 0.2$	83.1 $\uparrow 0.6$
+ Patch size 14	78.9 $\uparrow 0.2$	83.5 $\uparrow 0.4$
+ Teacher momentum 0.994	79.4 $\uparrow 0.5$	83.6 $\uparrow 0.1$
+ Tweak warmup schedules	80.5 $\uparrow 1.1$	83.8 $\uparrow 0.2$
+ Batch size 3k	81.7 $\uparrow 1.2$	84.7 $\uparrow 0.9$
+ Sinkhorn-Knopp	81.7 =	84.7 =
+ Untying heads = DINOv2	82.0 $\uparrow 0.3$	84.5 $\downarrow 0.2$

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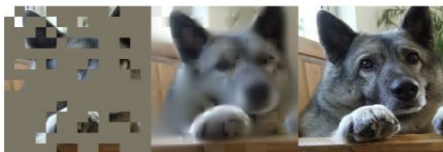
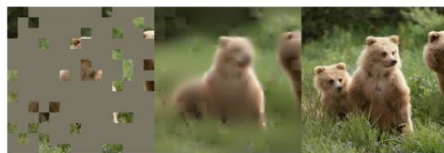
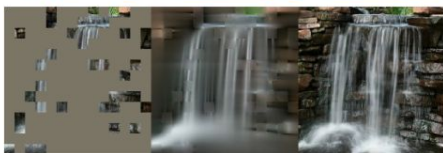
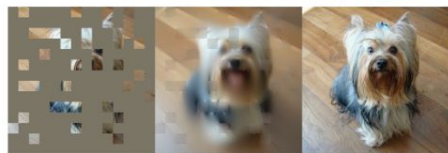
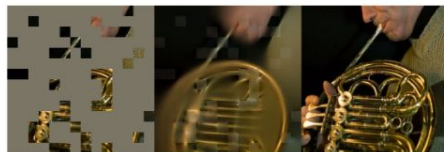
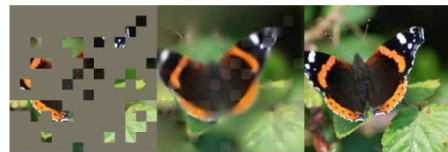
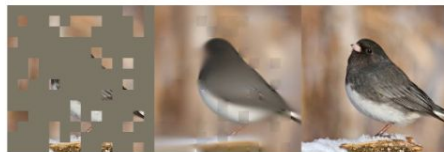
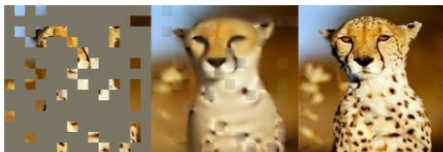


Masked Autoencoders (MaE)

A simple self-supervised architecture, easy to use, with few hyper parameters.



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Discuss: BERT is trained with cross entropy loss. What loss function should we use for MaE?

MaE Results

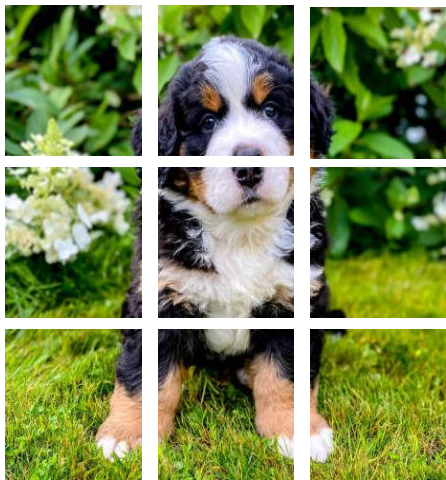
- Compared to supervised ViTs
 - Requires minimal data augmentation
 - Transfers better to downstream vision tasks
 - Object detection, segmentation

case	ft	lin
none	84.0	65.7
crop, fixed size	84.7	73.1
crop, rand size	84.9	73.5
crop + color jit	84.3	71.9

(e) **Data augmentation.** Our MAE works with minimal or no augmentation.

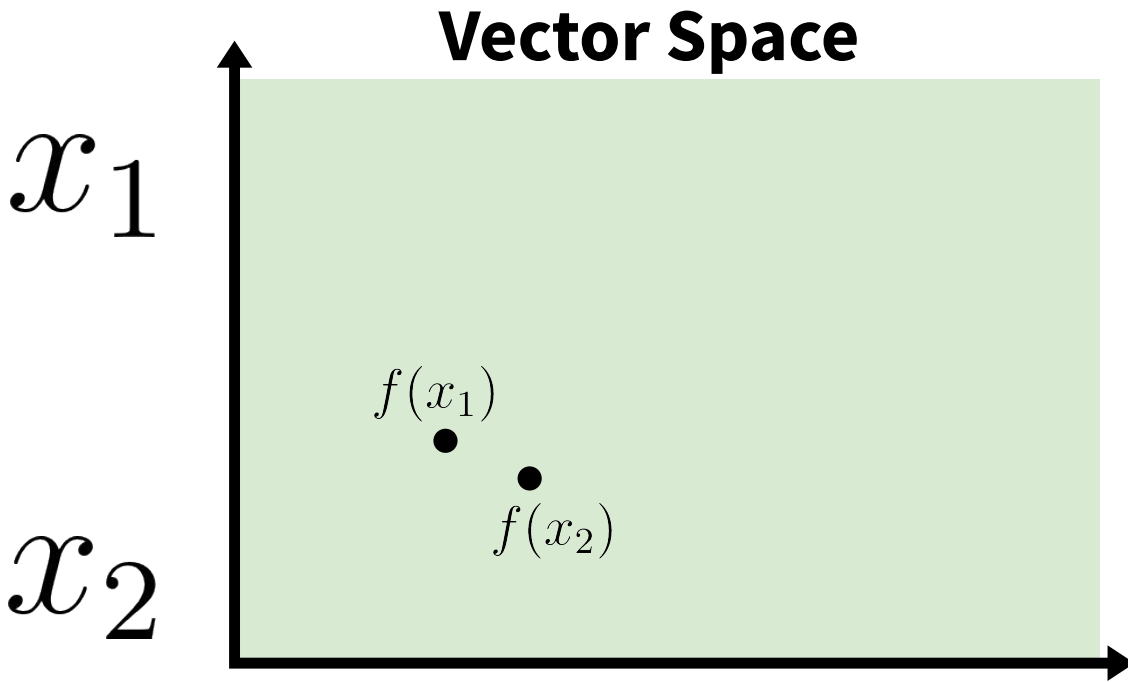
method	pre-train data	AP ^{box}		AP ^{mask}	
		ViT-B	ViT-L	ViT-B	ViT-L
supervised	IN1K w/ labels	47.9	49.3	42.9	43.9
MoCo v3	IN1K	47.9	49.3	42.7	44.0
BEiT	IN1K+DALLE	49.8	53.3	44.4	47.1
MAE	IN1K	50.3	53.3	44.9	47.2

Table 4. **COCO object detection and segmentation** using a ViT Mask R-CNN baseline. All entries are based on our implementation. Self-supervised entries use IN1K data *without* labels. Mask AP follows a similar trend as box AP.

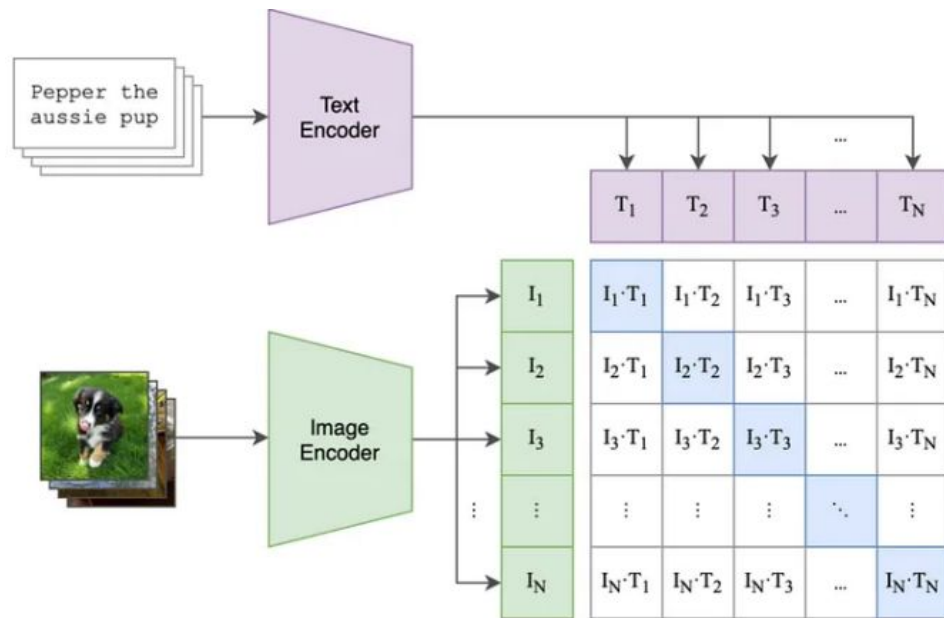


A puppy sits
with flowers

$f(x)$ = transformer rep.



CLIP (Contrastive Language-Image Pre-training)



Discuss: How can you train this model?

```
# image_encoder - ResNet or Vision Transformer
# text_encoder  - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l]       - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t             - learned temperature parameter

# extract feature representations of each modality
I_e = image_encoder(I) # [n, d_i]
```

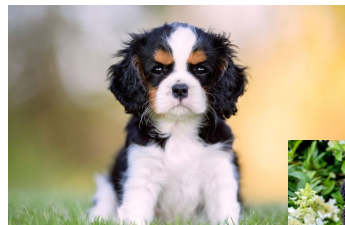
- Trained on 256 V100 GPUs for two weeks on 400 million (image, text pairs)
- On AWS, this would cost at least 200k dollars

```
# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

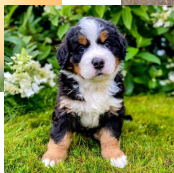
# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss   = (loss_i + loss_t)/2
```

Ranking using CLIP

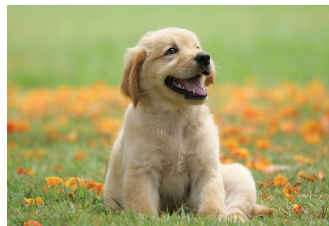
sim score - 0.9



sim score - 0.3



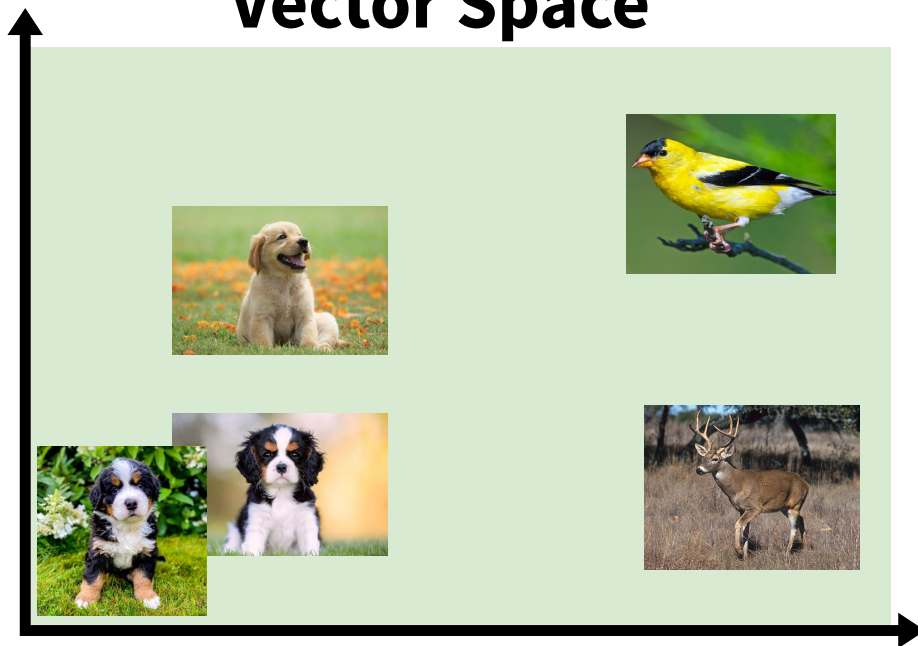
sim score - 0.6



sim score - 0.1



Vector Space

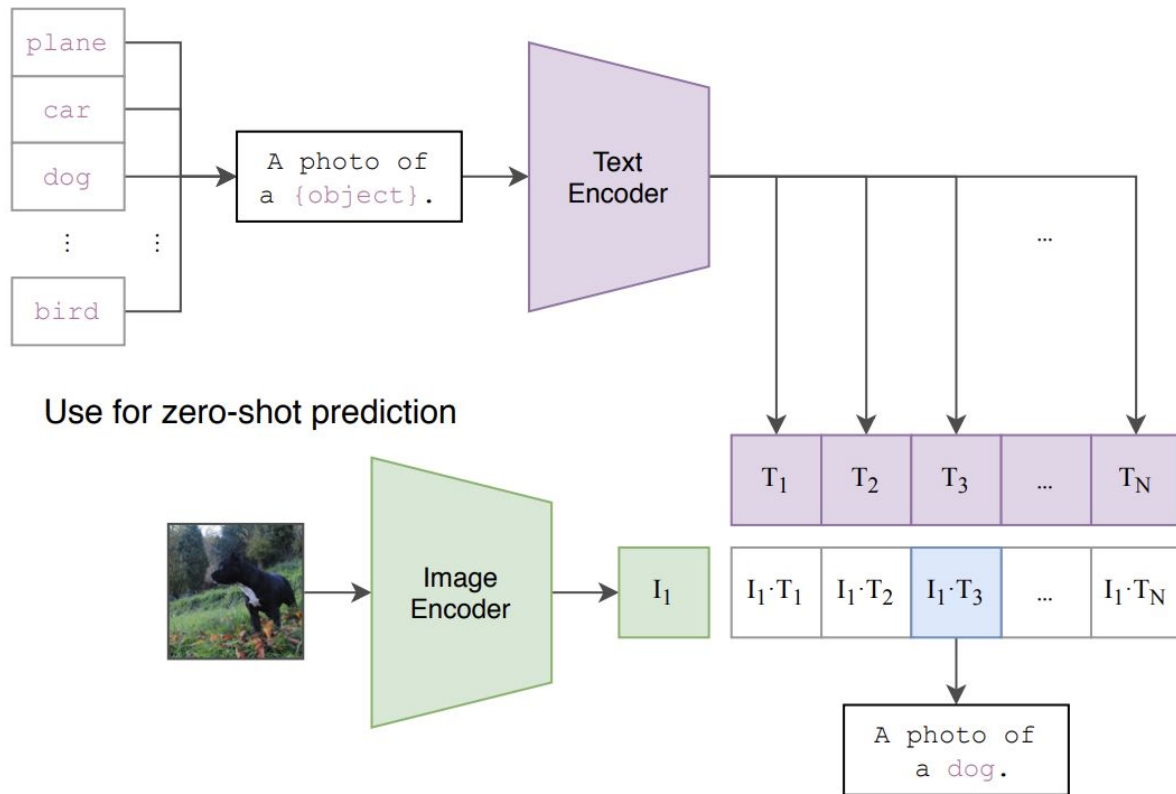


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Clip demo

<https://huggingface.co/spaces/vivien/clip>

Create dataset classifier from label text



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FOOD101

guacamole (90.1%) Ranked 1 out of 101 labels



✓ a photo of **guacamole**, a type of food.

✗ a photo of **ceviche**, a type of food.

✗ a photo of **edamame**, a type of food.

✗ a photo of **tuna tartare**, a type of food.

✗ a photo of **hummus**, a type of food.

YOUTUBE-BB

airplane, person (89.0%) Ranked 1 out of 23



✓ a photo of a **airplane**.

✗ a photo of a **bird**.

✗ a photo of a **bear**.

✗ a photo of a **giraffe**.

✗ a photo of a **car**.

SUN397

television studio (90.2%) Ranked 1 out of 397



✓ a photo of a **television studio**.

✗ a photo of a **podium indoor**.

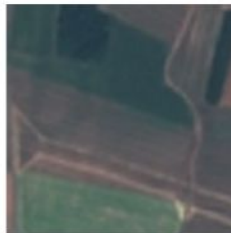
✗ a photo of a **conference room**.

✗ a photo of a **lecture room**.

✗ a photo of a **control room**.

EUROSAT

annual crop land (12.9%) Ranked 4 out of 10



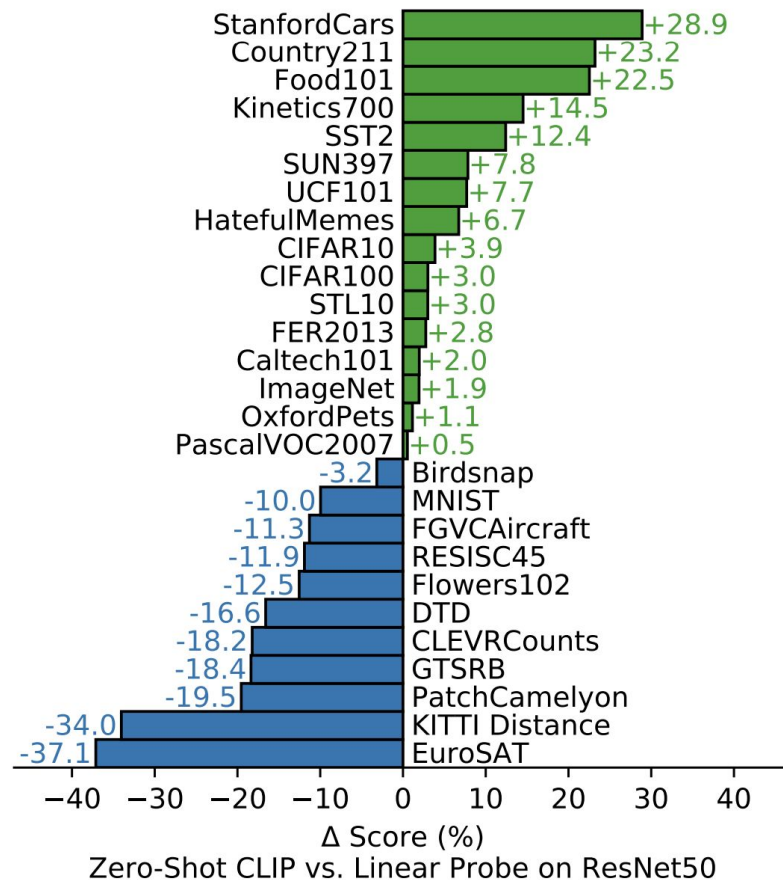
✗ a centered satellite photo of **permanent crop land**.

✗ a centered satellite photo of **pasture land**.

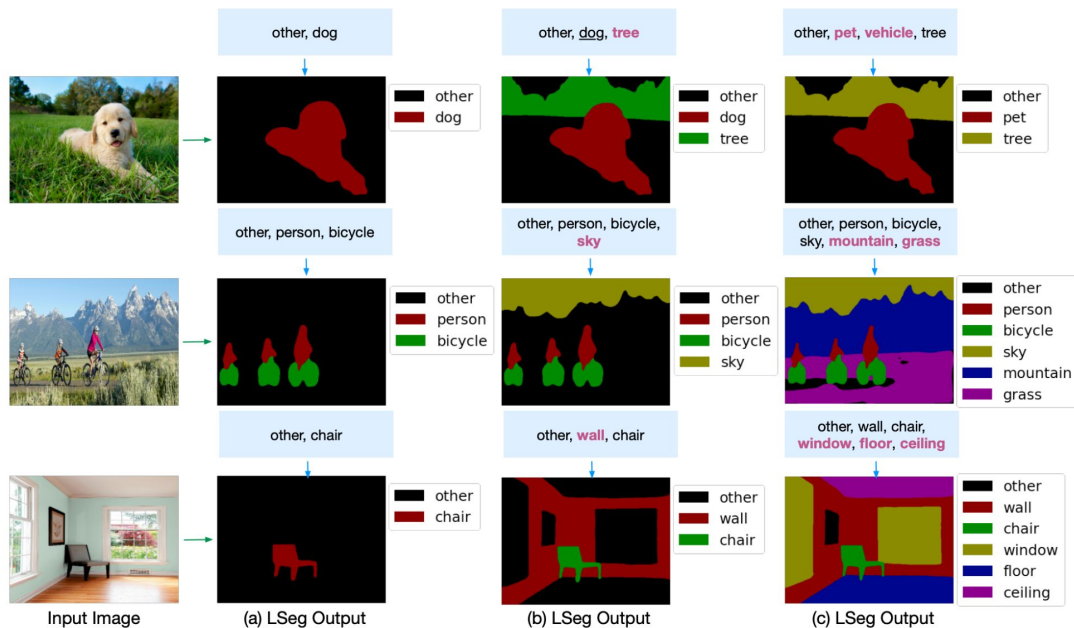
✗ a centered satellite photo of **highway or road**.

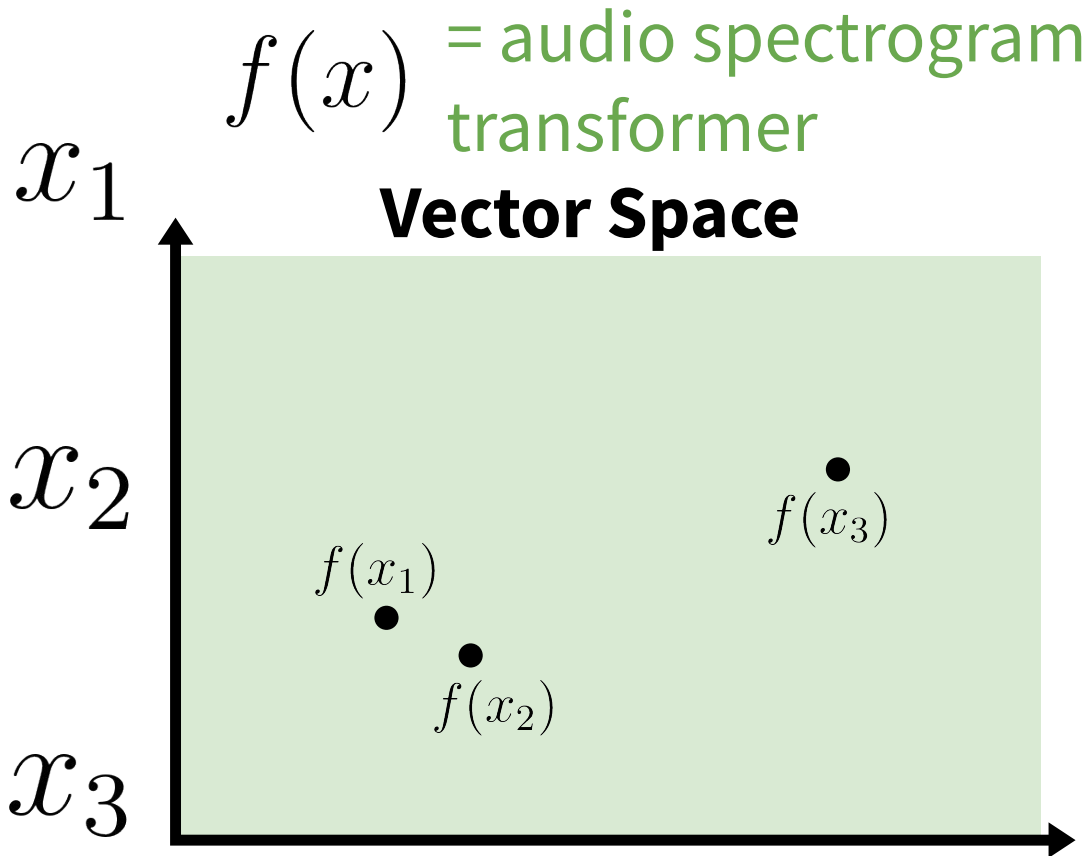
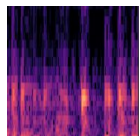
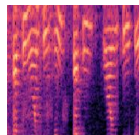
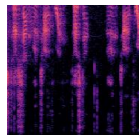
✓ a centered satellite photo of **annual crop land**.

✗ a centered satellite photo of **brushland or shrubland**.

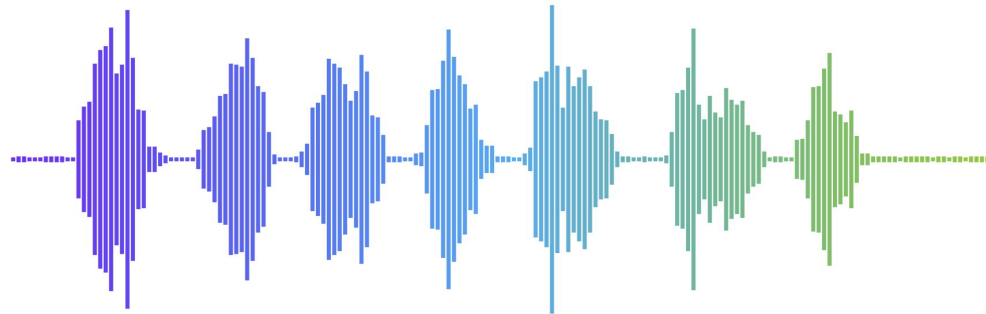


Application of CLIP





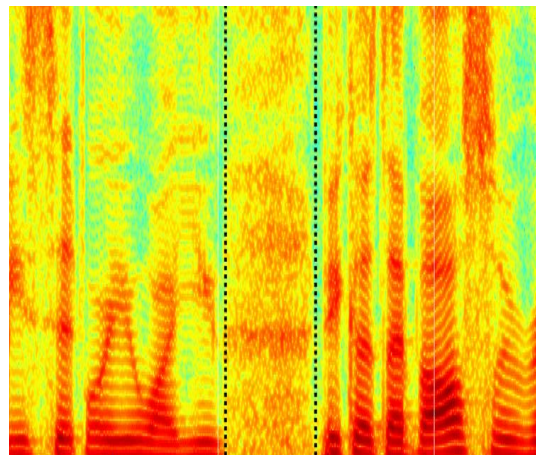
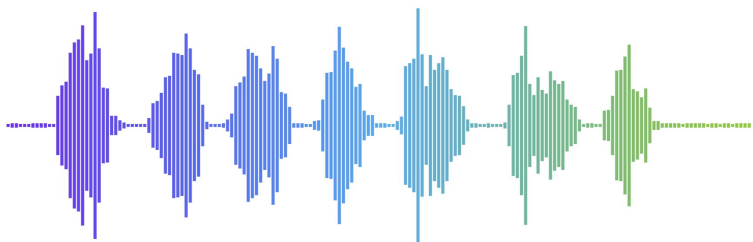
Audio Processing



Audio File

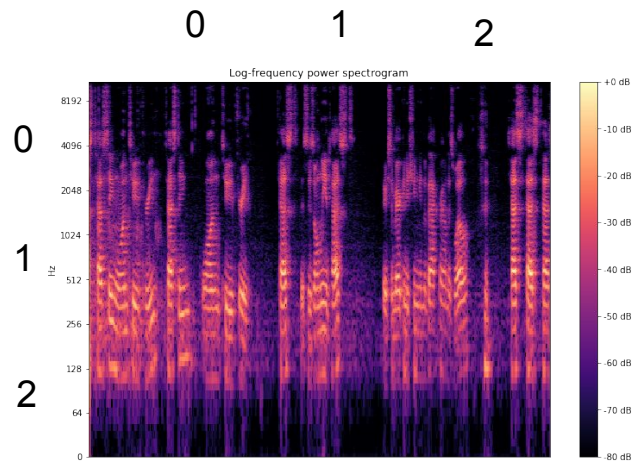
Spectrogram:

- Energy, pitch, fundamental frequency
- Decomposes signal into frequencies and their corresponding amplitudes

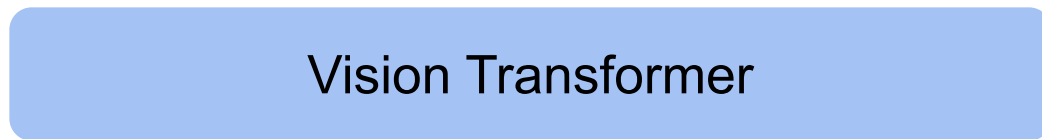


Spectrogram source: [Dumpala 2017](#)

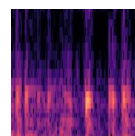
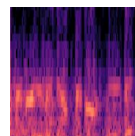
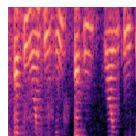
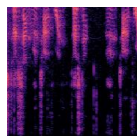
Audio as a vision problem



Most likely word sequence



Vision Transformer



...



$\langle(0,0)\rangle$

$\langle(0,1)\rangle$

$\langle(0,2)\rangle$

$\langle(1,1)\rangle$

Review

- Transformers can be used for vision tasks
 - Typically do not use convolution or other image specific architectures - just flattened image patches
 - Consequently they require more data
 - However they are very flexible and can be used with other data modalities
 - Learn positional embedding
- Swin transformers
 - Learn features at different scales
 - Scale linearly with the image size
 - Are less popular because of their complexity
- Self-supervised learning
 - Provides Transformer based models with the amount of data they need
 - Often used as backbone pre-trained models
 - Dino and MAE both learn very good embeddings
- Multi-modality
 - CLIP learns a common embedding for images and text
 - Can be used to retrieve images or parts of images from plain text